Felix Zhang

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Education

University of Toronto Sept. 2025 – June 2027
Masters of Science in Computer Science

University of Toronto Sept. 2021 – June 2025
Honours Bachelor of Science in Computer Science; Major in Mathematics *3.96/4.0 cGPA*

Publications

PD3: Prefetching Data with DPUs for Disaggregated Memory May 2026
Sidharth Sankhe, Felix Zhang, Umayrah Chonee, Sherman Lim, Jason Hu, Jialin Li, Qizhen Zhang
23rd USENIX Symposium on Networked Systems Design and Implementation

Research Experience

Research Assistant; Far Data Lab, University of Toronto Sept. 2024 – Aug. 2024

- Investigated offloading computations onto SmartNICs and data processing unit (DPUs)
- Parallelized the execution of *Monodepth2* in **Python** and **C++** with the latter achieving linear performance scaling with the number of threads
- Built a DPU-based prefetcher *PD3* with a team of 6 which intercepts network traffic to predict and prefetch data for tiered key-value stores
- Designed an external service for offloading shuffle operations from database management systems which supports either a memory or storage backend managed by DPUs

Research Assistant; University of Toronto March 2024 – Dec. 2024

- Worked on a team of 11 to implement a pedagogical kernel *KidneyOS* in **Rust** to be used in an introductory operating systems course with **500+** students annually
- Enabled thread creation and destruction, multi-threading, pre-emptive scheduling within the thread system
- Led a team of 3 to implement POSIX-compatible syscalls and add support running user-space executables

Industry Experience

ML Runtime Engineer; Cerebras Systems May 2024 – Aug. 2025

- Implemented a runtime virtual memory system in **C++** with a team of 3 which pre-emptively loads data before it is accessed, allowing off-chip memory to be used for the first time with only a **10%** performance penalty
- Added support for network storage in the paging system with remote direct memory access, providing **100** GB/s read and write speeds with **10** μs latency to multiple remote servers
- Enabled the ability log and replay the network operations, decreasing the time to recreate stalls and timeouts by over **80%**, and setup unit tests to automatically catch breakages and performance regressions in the network layer
- Improved the throughput of the network layer by **6%** when transferring data by implementing best practices for remote direct memory access and reducing setup overhead
- Determined the cable and port mapping for a one node cluster which will be used for all future deployments

Awards & Scholarships

Ontario Graduate Scholarship ($15 000), University of Toronto May 2025 – April 2026
Queen Elizabeth II Graduate Scholarships in Science and Technology ($15 000) (Declined), University of Toronto May 2025 – April 2026
Fields Undergraduate Summer Research Program ($3 800), Fields Institute June – Aug. 2024
Louis Savlov Scholarship in Sciences And Humanities ($1 000), University of Toronto Nov. 2023 – Jan. 2025
Dean's List Scholar, University of Toronto Jan. 2022 – June 2025
Second Malcom Wallace Scholarship ($5 000), University of Toronto Sept. 2021 – Oct. 2024
University of Toronto Scholar ($7 500), University of Toronto Sept. 2021

Technical Skills

Languages C++, Rust, Python, C, C#, Bash, Java
Frameworks Catch2, PyTest, PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, SciPy, Eigen3
Tools Git, Linux, Unix, tmux, CMake, Docker, Anaconda, Google Colab, QEMU, Jupyter, WSL, Slurm